

You are building a comprehensive UX simulation document for H.O.N. — a near-launch iOS workout tracking app. Your output is a single self-contained HTML file. Read every word of this prompt before writing a single line of code.

PART 1: APP CONTEXT

What H.O.N. is

H.O.N. (Habit Over Numbers) is an iOS workout tracking app near launch. Broad audience: casual lifters, serious gym-goers, beginners.

Core workout loop: Start session → select exercise → log sets (weight + reps) → rest timer → post-session emoji/feel selector → AI-generated feedback phrase → session summary

Key metric: e1RM trajectory as a rate-of-change signal. Derived metrics: rate of strength gain (kg/week, %/week), relative strength ($\text{e1RM} \div \text{bodyweight}$), volume load, INOL.

Phrase bank: AI-generated post-session feedback. Structure: [Session Frame] + [What Happened] + [Trend Modifier] + [What's Next] + [Connective Tissue]. Max 3 sentences / 350 characters. Neutral tone. Four dimensions: Reps outcome (4 tiers), Weight state (5 states), Trend (5 types, omitted if <3 sessions), History (4 types).

UX constraint: 3 objective inputs per set (weight, reps, set timer) + one post-session emoji/feel selector. Minimal cognitive load.

Design system — locked, do not deviate:

```
--onyx: #111111      primary background
--carbon: #1c1c1e     card backgrounds
--iron: #2c2c2e       borders, dividers
--iron-light: #3a3a3c subtle borders
--amber: #d4943a      primary accent, hero metric ONLY
--ivory: #f5f0e8      primary text
--ivory-dim: rgba(245,240,232,0.6)
--ivory-muted: rgba(245,240,232,0.35)
--sage: #7a9e8e
--slate: #6b85a0
--rose: #b47a80
--clay: #a07060
--lavender: #8a80a8
```

Fonts: Load from Google Fonts.

- DM Serif Display — headings, display text, persona names
 - DM Mono — labels, metadata, code, timestamps, badge text
 - DM Sans — body, narrative text, annotations
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PART 2: THE FORMULA COMPARISON PROTOCOL

This applies to EVERY persona and EVERY test case that involves a calculated metric. It is not optional.

The assumption

H.O.N. uses the Epley formula to calculate e1RM:

$$e1RM = weight \times (1 + reps / 30)$$

Derived metrics assumed to be:

- Rate of strength gain = (most recent e1RM – earliest e1RM) ÷ number of weeks elapsed
- Relative strength = e1RM ÷ bodyweight
- Volume load = sum of (weight × reps) across all sets in a session

What you must do in every formula-touching test case

Whenever a persona logs a set that triggers a metric calculation, the simulation document must show THREE things side by side:

1. **Assumed formula result** — what the Epley formula above would produce with those inputs
2. **Observed app result** — a field the developer fills in by actually checking the app
3. **Delta analysis** — a written interpretation: does the difference make sense? Could it be a different valid formula? Could it be a bug? What are the implications?

Format this as a “Formula Audit Block” — a distinct styled card within the annotation, separate from the narrative. It must include:

- The inputs used (weight, reps)

- The assumed formula written out with the actual numbers substituted
- The expected output
- A blank "App shows:" field for the developer to fill in
- A "Delta:" field with possible explanations pre-written (e.g. "If app shows 71.2kg, it may be using Brzycki. If app shows 68kg, check for rounding. If app shows anything above 80kg with these inputs, it is a calculation error.")

Common formula variants to account for

Instruct the developer to check against these alternatives when a discrepancy appears:

- **Epley:** $\text{weight} \times (1 + \text{reps}/30)$ — assumed
- **Brzycki:** $\text{weight} \times (36 / (37 - \text{reps}))$ — common alternative, diverges significantly above 10 reps
- **Lander:** $(100 \times \text{weight}) / (101.3 - 2.67123 \times \text{reps})$
- **Lombardi:** $\text{weight} \times \text{reps}^{0.10}$

For any discrepancy, the delta analysis should tell the developer: which formula would produce the observed number, whether that formula is a legitimate choice, and whether the choice creates misleading output at edge cases (e.g. Brzycki breaks down above 36 reps; Epley produces absurd values above ~30 reps without a rep cap).

PART 3: THE PERSONAS

There are 12 personas total. For each one you must:

1. Write a persona card (emoji, name, age, tagline, bio paragraph, trait pills)
2. Simulate 5 phases of their day (Morning Glance, Pre-Workout, Active Session, Post-Workout, Evening)
3. Generate **at minimum 4 distinct test cases per phase** — not just narrative moments, but explicitly labeled TEST CASES with: a test name, the exact scenario, what to observe, what pass looks like, what fail looks like
4. Apply formula audit blocks wherever a calculation is involved
5. Write a scorecard with friction count, feeling count, interruption count, critical path

count, and a prioritized fix list of 4–6 items

6. End each persona with one “Design Question” — the single question this persona forces the developer to answer before shipping

Annotation types — use consistently throughout

Every narrative moment gets at least one annotation. Types:

- **Friction** (red) — something that breaks flow Style: `background: rgba(180,60,60,0.15); border: 1px solid rgba(180,60,60,0.5); color for label: #e07070`
 - **Feeling** (amber) — emotional design moment Style: `background: rgba(212,148,58,0.12); border: 1px solid rgba(212,148,58,0.4); color for label: #d4943a`
 - **Interruption** (slate) — real-world edge case Style: `background: rgba(107,133,160,0.15); border: 1px solid rgba(107,133,160,0.45); color for label: #90b0cc`
 - **Flow Check** (sage) — pass/verify step Style: `background: rgba(122,158,142,0.1); border: 1px solid rgba(122,158,142,0.35); color for label: #7a9e8e`
 - **Formula Audit** (lavender) — calculation verification Style: `background: rgba(138,128,168,0.12); border: 1px solid rgba(138,128,168,0.4); color for label: #8a80a8`
-

PERSONA 1: Marcus, 34 — The Inconsistent Committer

Tagline: THE WELL-INTENTIONED GAPPER **Avatar:**  **Bio:** Works from home, two kids under 6. Wants to lift 4x/week, manages 2–3. Home gym in the garage. Phone always nearby. Gets interrupted constantly. Intermediate lifter with opinions about form but inconsistent enough that progress feels invisible. **Traits:** home gym · distracted · inconsistent · 2–3x/week · intermediate · two kids

Core tension: The app lives inside a chaotic day. Session resumability, home screen context after gaps, and post-session phrase quality are all critical.

Test cases to generate (at minimum 4 per phase):

Phase 1 — Morning Glance (7am, in bed):

- TC1.1: Opens app after 4-day gap. What does home screen show? Does it surface the

gap at all?

- TC1.2: Taps into last session to check bench numbers. Count taps from home screen. Record time.
- TC1.3: Sees his last e1RM. Apply formula audit: does the displayed number match Epley for his last logged set?
- TC1.4: Closes app in 8 seconds having gotten nothing useful. What was missing?
- TC1.5: Tries to signal intent to train today without starting a session. Is there any way to do this?

Phase 2 — Pre-Workout (11:50am, garage):

- TC2.1: Opens app one-handed while loading the barbell. Which inputs require two hands?
- TC2.2: Tries to start a session while phone is in his pocket and pulls it out mid-task. Does screen state make sense?
- TC2.3: Searches for "bench press" — how many taps to get to the exercise and have it ready to log?
- TC2.4: Checks if rest timer default is appropriate for his training style. Can he change it?

Phase 3 — Active Session (12:02–12:48pm):

- TC3.1: Logs 80kg × 6. Formula audit block required. App shows e1RM: _____. Expected (Epley): $80 \times (1 + 6/30) = 96\text{kg}$. Delta: if app shows 95.2kg, check Brzycki. If app shows 98kg+, possible bug.
- TC3.2: Screen locks mid-rest-timer. Returns. Does timer show correct remaining time or does it reset?
- TC3.3: Kid interrupts. App backgrounds for 6 minutes. Returns. Session state audit: are logged sets intact, is session still active, is timer state preserved?
- TC3.4: Logs wrong weight (60 instead of 70). Realizes immediately. Steps to correct: count taps, measure time.
- TC3.5: Rapid tap test — taps "log set" 8 times quickly without changing values. How many sets appear?
- TC3.6: Logs sets for 3 different exercises. Wants to go back to first exercise. Is there a way to navigate between exercises in-session?

Phase 4 — Post-Workout (12:48pm, done):

- TC4.1: Ends session with sweaty hands. Tap targets audit — which buttons are hit on first try, which require multiple attempts?
- TC4.2: Feel selector appears. He selects “strong.” Does this visibly affect anything in the session summary or feedback phrase?
- TC4.3: Feedback phrase fires. Rate it: does it reference his specific data (weight, reps, gap since last session)?
- TC4.4: Force-quits app immediately after tapping “end session” before summary loads. Reopens. Is session saved?

Phase 5 — Evening (9:30pm, couch):

- TC5.1: Opens app with no intention to log. What is there to look at? Time how long before he closes it.
- TC5.2: Checks bench press trend over last 4 sessions. Formula audit: are the e1RM values on the chart consistent with Epley for those logged sets?
- TC5.3: Tries to find his rate of strength gain metric. Does it exist? Is it legible without explanation?
- TC5.4: Wonders what happens if he doesn’t train tomorrow. Does the app have any opinion on this?

Design Question for Marcus: When a user hasn’t trained in 4 days and opens the app at 7am, what is the single most useful thing H.O.N. can show them — and does it currently show that?

PERSONA 2: Priya, 29 — The Analytical Lifter

Tagline: THE SPREADSHEET REFUGEE **Avatar:** 🎧 **Bio:** Software engineer, no kids, gym 5 min away. Trains 4–5x/week at 6am. Analytical and data-driven. Currently tracking on a spreadsheet because no app has been precise enough. Skeptical of AI feedback but would be won over by accuracy. Has been burned by “motivational” apps before. **Traits:** commercial gym · 4–5x/week · data-driven · early morning · skeptic · spreadsheet refugee

Core tension: Every number H.O.N. shows must be defensible. The feedback phrase is her credibility test.

Test cases (at minimum 4 per phase):

Phase 1 — Morning Glance (5:40am):

- TC1.1: Opens app to check last push session weights. How many taps? What's the fastest path?
- TC1.2: Wants to see her last bench press specifically, not the whole session. Exercise-level drill-down tap count.
- TC1.3: Checks her e1RM from last session. Formula audit required: expected vs. observed.
- TC1.4: Tries to identify what she should attempt today based on last session data. Does the app suggest anything, or only record?
- TC1.5: Notices she has 3 sessions this week. Does the app surface weekly volume anywhere she can see in under 10 seconds?

Phase 2 — At the Gym (6:05am):

- TC2.1: One-handed navigation while adjusting bench. Which session screens require two hands?
- TC2.2: Earphones in, music playing. Does any notification or alert from H.O.N. interrupt her audio?
- TC2.3: Exercise selection — she wants "Barbell Bench Press" not "Bench Press (Machine)." Are these disambiguated clearly?
- TC2.4: Someone asks if she's using the equipment. She pauses. Rest timer runs unattended. What state does she return to?

Phase 3 — Active Session:

- TC3.1: Attempts 60kg × 4 (weight jump from 57.5kg, rep drop from 5). Formula audit: expected e1RM = $60 \times (1 + 4/30) = 68\text{kg}$ vs. previous $57.5 \times (1 + 5/30) = 67.1\text{kg}$. App shows: _____. Is the progression correctly read as marginal improvement?
- TC3.2: Logs 5 exercises, 4 sets each (20 total sets). Does session performance degrade (slower inputs, UI lag) by set 15+?
- TC3.3: Wants to log a "notes" field for a set — e.g., "paused reps." Is this possible? If not, how does she record qualitative detail?
- TC3.4: Notices rest timer default doesn't match her actual rest periods. Can she change it mid-session? Does it remember her preference?
- TC3.5: Logs a warmup set she doesn't want counted in her e1RM. Is there a way to mark a set as warmup/exclude from calculation?
- TC3.6: Airplane mode test — gym basement has no signal. Session must continue to

function and save locally.

Phase 4 — Post-Session (subway, 7:15am):

- TC4.1: Reviews session summary on subway. Finds a missed set of rows. Can she add it retroactively? Steps to do so.
- TC4.2: Feedback phrase fires. She reads it analytically. Does it mention the weight jump? The rep drop? The e1RM comparison to last session?
- TC4.3: She tries to calculate whether today was a PR. Does H.O.N. indicate PRs? How?
- TC4.4: Session summary screenshot test — does it look clean and legible in iMessage?

Phase 5 — Evening Analysis (7:45pm):

- TC5.1: Navigates to bench press history. Tap count from home screen to 6-week e1RM trend.
- TC5.2: Formula audit on trend line: pick 3 sessions from the chart and verify the e1RM values match Epley for the underlying logged sets.
- TC5.3: Checks rate of strength gain. Formula audit: $\text{expected} = (\text{most recent e1RM} - \text{earliest e1RM}) \div \text{weeks}$. App shows: _____. Delta: does this match?
- TC5.4: Discovers her squat e1RM has been flat for 3 weeks. Does the app surface this, or does she have to find it herself?
- TC5.5: Tries to export her data. Is there any data export path? What format?

Design Question for Priya: If Priya's manual Epley calculation and H.O.N.'s displayed e1RM differ by more than 2kg — and there's no in-app explanation of which formula is used — does she stay or leave?

PERSONA 3: Danny, 23 — The Hopeful Beginner

Tagline: THE FIRST-TIMER **Avatar:** 🌱 **Bio:** Recent grad, 8 weeks into lifting, first gym membership. Long enough to feel less lost, not long enough to have real opinions. Intimidated by the gym floor. Goes 2–3x/week after work. Doesn't know what e1RM means. Prone to quitting if the app makes him feel dumb. **Traits:** beginner · 8 weeks in · commercial gym · after work · intimidated · no baseline

Core tension: The app's complexity must not outpace Danny's experience level. Every unfamiliar metric is a potential exit point.

Test cases (at minimum 4 per phase):

Phase 1 — Morning Glance (commute):

- TC1.1: Opens app hoping it will motivate him to go tonight. What does the home screen offer?
- TC1.2: Sees a metric he doesn't recognize (e1RM). Is there any explanation available without leaving the app?
- TC1.3: His last session was 3 days ago. Does the app feel like it missed him or doesn't care?
- TC1.4: He taps around exploring. Identify every screen that uses terminology a beginner wouldn't know. List them.

Phase 2 — At the Gym (6:30pm):

- TC2.1: Stands in front of squat rack. Opens H.O.N. to figure out what to do. Does the app help or just wait?
- TC2.2: Searches "cable" — gets 8 options. Does he have enough information to pick the right one?
- TC2.3: Starts a session. How many screens does he encounter before he can log his first set?
- TC2.4: Accidentally starts a second session while one is already active. What happens?

Phase 3 — Active Session:

- TC3.1: Logs squat 40kg × 8. Formula audit: $\text{expected e1RM} = 40 \times (1 + 8/30) = 50.7\text{kg}$. App shows: _____. Does this number mean anything to him without context?
- TC3.2: Logs exercise under wrong name ("squat" vs. "barbell back squat"). This fragments history. Is there any disambiguation prompt?
- TC3.3: Forgets to log one set in the middle of his session. Logs the next exercise. Comes back. Can he insert a set retroactively within an active session?
- TC3.4: Rest timer fires an alert while he's mid-conversation with someone at the gym. Is the alert discreet or embarrassing?
- TC3.5: He logs 1 rep at 60kg as a test — he just wanted to see what happened. Formula audit: $\text{Epley with 1 rep} = 60 \times (1 + 1/30) = 62\text{kg}$. Is this a meaningful number? Does the app treat it the same as a real working set?

Phase 4 — Post-Workout (leaving gym, 7:35pm):

- TC4.1: Feedback phrase fires. He has 3 sessions logged — just enough for a trend. Does the phrase reference the trend? Is it specific to his 40kg squat?
- TC4.2: He wants to screenshot and send to a friend. Session summary screenshot audit — does it look impressive or like a data dump?
- TC4.3: He accidentally skips the feel selector by tapping too fast. Is there a way to go back and add it?
- TC4.4: He closes the app without reading the feedback phrase. Next time he opens it, is the phrase still accessible?

Phase 5 — Late Night (10pm, in bed):

- TC5.1: Looks at his squat history — 3 sessions logged. Does the chart convey progress with only 3 points?
- TC5.2: Formula audit on those 3 data points: verify each plotted e1RM against Epley for the underlying sets.
- TC5.3: He doesn't understand what the y-axis on the chart means. Is it labeled? Is the unit clear?
- TC5.4: He wants to know if he's doing well for someone 8 weeks in. Does the app have any way to answer this?

Design Question for Danny: At 3 sessions logged, what is the minimum viable "progress" signal the app can give him — and is that signal currently present?

PERSONA 4: Wei, 31 — The Returner

Tagline: THE COMEBACK **Avatar:**  **Bio:** Trained consistently for 4 months, stopped for 3 weeks (holiday, then illness). Now returning. His data exists but feels like ancient history. Anxious about how far he's regressed. Doesn't want the app to confront him with his past bests. **Traits:** returning user · 3-week gap · previously consistent · anxiety about regression · intermediate · home gym

Core tension: Re-entry tone. The app's emotional stance toward someone who's been gone is a product decision.

Test cases (at minimum 4 per phase):

Phase 1 — First open after 3-week gap:

- TC1.1: Home screen audit — does it surface the 3-week gap? What exact language or

visual does it use?

- TC1.2: He sees his previous e1RM on bench press: 92kg. Formula audit: verify this is consistent with his last logged set. Today he expects to hit ~80kg. Does the app set an implicit expectation?
- TC1.3: He tries to find a way to tell the app "I've been sick, reset expectations." Is this possible?
- TC1.4: His trend line has a visible cliff at week 4 (when he stopped). How does the visualization handle a 3-week gap in data — flat line, broken line, gap, or nothing?
- TC1.5: He opens the app 3 times in 10 minutes, anxiously. Does anything change? Is there any state that responds to repeated opens?

Phase 2 — First session back:

- TC2.1: He logs bench 75kg × 4 (down from 85kg × 5 three weeks ago). Formula audit: expected e1RM = $75 \times (1 + 4/30) = 85\text{kg}$ vs. previous $85 \times (1 + 5/30) = 99.2\text{kg}$. App shows: _____. The drop is significant — how does the trend line absorb this?
- TC2.2: Feedback phrase fires on a session with lower weights than his history. Does it acknowledge the return context? Or treat this as a straightforward decline?
- TC2.3: He finishes a lighter-than-usual session. The feel selector appears. He taps "strong" — because he does feel good about being back. Does "strong" affect the phrase interpretation even when the numbers are down?
- TC2.4: He sees his new e1RM plotted on the chart. The dip is visible. Does the visualization make this feel like information or like failure?

Phase 3 — Second session (2 days later):

- TC3.1: He logs bench 78kg × 5. Formula audit: expected e1RM = $78 \times (1 + 5/30) = 91\text{kg}$. That's nearly back to his previous level in 2 sessions. Does the app recognize rapid recovery? Does the rate-of-gain metric spike in a way that's meaningful?
- TC3.2: Trend is now: [previous plateau] → [big dip] → [rapid recovery]. Does the trend algorithm correctly identify this shape? What does the feedback phrase call it?
- TC3.3: He compares his current e1RM to his pre-gap e1RM. Is this comparison surfaced anywhere, or does he have to scroll through history to do it manually?
- TC3.4: He wants to mark the gap period with context ("was sick"). Is there any journaling, annotation, or session-note capability?

Phase 4 — Evening reflection:

- TC4.1: He's relieved the comeback is going well. Does the app reflect this at all — any "you've returned" framing?
- TC4.2: He browses his full history including the gap. The gap is visible. Does he feel informed or judged?
- TC4.3: Rate of strength gain formula audit: with the dip-and-recovery, what does the algorithm calculate? Does it smooth, ignore the gap, or count it? Expected behavior vs. observed.

Design Question for Wei: When a user's first session back is weaker than their historical average, does the feedback phrase know they're returning — or does it treat their regression as a setback?

PERSONA 5: Serena, 27 — The Phone-Averse Lifter

Tagline: THE RETROACTIVE LOGGER **Avatar:**  **Bio:** Competitive powerlifter, 4x/week, strict no-phone-on-the-floor rule. Logs everything from memory or a paper notepad, 20–60 min after the session. H.O.N. was designed for in-session use. Serena is using it in a mode that wasn't designed for her. **Traits:** powerlifter · 4x/week · no phone during training · retroactive logger · paper notes · high data precision

Core tension: Is H.O.N. usable as a post-session logging tool, or is it fundamentally in-session only?

Test cases (at minimum 4 per phase):

Phase 1 — Home, 45 min after session:

- TC1.1: Opens H.O.N. 45 minutes after finishing a session. Starts a session. The "session" technically started now, not 45 min ago. Can she correct the start time?
- TC1.2: She needs to log 5 exercises × 4 sets = 20 sets from memory. Is there any efficiency mode for bulk entry? How long does this actually take?
- TC1.3: She enters her sets out of order — she remembers deadlifts last even though they were first. Is exercise order editable?
- TC1.4: Rest timer is running and prompting her to log. She doesn't need it. Is there a way to dismiss it permanently for this session?

Phase 2 — Bulk retroactive entry:

- TC2.1: Her notes say "3 × 5 @ 92.5kg." She logs 92.5kg — does the app accept decimals

for weight? Test with 92.5, 77.5, 102.5.

- TC2.2: Her notes say "1 × 3 @ 100kg (failed rep 4)." She wants to log a failed/incomplete set. Is this possible? Options: log 3 reps, log 4 reps, or some incomplete set marker?
- TC2.3: Formula audit on 92.5kg × 5: expected e1RM = $92.5 \times (1 + 5/30) = 108\text{kg}$. App shows: _____. For a powerlifter, precision matters — is the formula appropriate for low-rep, high-weight sets?
- TC2.4: She realizes she logged the same exercise twice (entered it as both "Deadlift" and "Conventional Deadlift"). Can she merge or delete the duplicate?
- TC2.5: She wants to log 4 sets of the same exercise quickly without navigating away between each. What is the fastest flow for this?

Phase 3 — Formula precision for powerlifting:

- TC3.1: Powerlifting formula audit — at low reps (1–5), different formulas diverge significantly. For 100kg × 1: Epley = 103.3kg, Brzycki = 102.7kg. For 100kg × 3: Epley = 110kg, Brzycki = 109.1kg. For 100kg × 5: Epley = 116.7kg, Brzycki = 117.6kg. App shows for each: _____. Which formula does H.O.N. use, and is it appropriate for powerlifting-range reps?
- TC3.2: She logs a 1-rep max attempt. Epley with 1 rep: $\text{weight} \times (1 + 1/30) = \text{weight} \times 1.033$. Is this meaningful? How does the app treat a logged 1RM vs. a calculated 1RM?
- TC3.3: She logs a competition-style session (10 total sets, 1–3 reps each, very high weight). Does the session summary look coherent? Does volume load calculate correctly?

Phase 4 — Post-entry review:

- TC4.1: She reviews her retroactively logged session. Does anything in the UI indicate when the session was logged vs. when it occurred?
- TC4.2: Feedback phrase fires. Her session was 45 minutes ago. Does the phrase feel relevant? Does it know she's in a strength-focused program (implied by her rep ranges)?
- TC4.3: She compares H.O.N.'s e1RM to her actual competition maxes. Formula audit: are they consistent? Should they be?

Phase 5 — Long-term use pattern:

- TC5.1: After 4 weeks of retroactive logging, does her trend data look accurate? Formula audit on trend line — pick 5 plotted points and verify against Epley for the underlying sets.

- TC5.2: Rate of strength gain formula audit: with her specific rep ranges (1–5), does the e1RM trajectory accurately represent her actual strength gains?

Design Question for Serena: Is H.O.N. a viable tool for someone who never uses their phone during training — or does the current UX make retroactive logging feel like a second-class experience?

PERSONA 6: James, 36 — The Plateau Fighter

Tagline: THE STUCK ONE **Avatar:** 🧔 **Bio:** 2 years of consistent lifting. 8 weeks stuck on the same bench numbers. Not injured — just stuck. Knows what a plateau is. Wants the app to tell him something he doesn't already know. **Traits:** experienced · 2 years lifting · 8-week plateau · analytical · frustrated · data-hungry

Core tension: H.O.N.'s value proposition collapses if it has nothing useful to say when progress is flat.

Test cases (at minimum 4 per phase):

Phase 1 — Morning review:

- TC1.1: Opens app. Bench e1RM trend is flat for 8 weeks. Formula audit: verify that “flat” is accurate — pick 3 sessions from the trend and verify the e1RM values match Epley.
- TC1.2: Rate of strength gain formula audit: $\text{expected} = (\text{week 8 e1RM} - \text{week 1 e1RM}) \div 8$. With a true plateau, this should be near-zero or negative. App shows: _____. Does it match?
- TC1.3: The trend visualization — is “flat for 8 weeks” visually obvious? Or does chart scaling make a small range look like noise?
- TC1.4: He tries to identify exactly when his progress last happened. Can he navigate history to find the last session where e1RM improved?

Phase 2 — Pre-workout:

- TC2.1: He wants to try a different rep scheme today to break the plateau. Does H.O.N. give him any basis for this decision, or only record it after?
- TC2.2: He checks his volume load over the past 8 weeks. Formula audit: $\text{volume} = \text{sum}(\text{weight} \times \text{reps})$ per session. Does the chart show this? Does it match manual calculation?
- TC2.3: He checks INOL if available. Is it visible? Is it labeled? Does the value (ideally 0.8–

1.2 for hypertrophy, >2 for overreaching) fall in a range that suggests anything?

Phase 3 — Session where he tries higher volume:

- TC3.1: Logs 75kg × 8 (higher reps, same weight). Formula audit: expected e1RM = $75 \times (1 + 8/30) = 95\text{kg}$. Compare to his previous 80kg × 5 e1RM = $80 \times (1 + 5/30) = 93.3\text{kg}$. App shows: _____. Does the different approach show up as an e1RM improvement even with the same “stuck” weight?
- TC3.2: Feedback phrase fires on a still-flat e1RM after 8 weeks. Does it acknowledge the plateau? Phrase audit: “Keep pushing!” = fail. “Your bench has held steady for 8 sessions — this kind of wall often responds to...” = pass direction.
- TC3.3: He changes his feel selector to “strong” despite the plateau. Does the phrase reconcile high feel with flat numbers?
- TC3.4: He logs a session with deliberately lower volume to test deload response. Does any metric reflect this?

Phase 4 — Evening pattern hunting:

- TC4.1: He’s looking for when his bench last progressed. Navigation test: how many taps to get to a session from 10 weeks ago?
- TC4.2: He wants to compare his squat trend (progressing) vs. bench trend (stalled) side by side. Is this possible in H.O.N.?
- TC4.3: Formula audit across exercises: do the e1RM trends for squat and bench use the same formula? Do they render at the same scale?

Design Question for James: When H.O.N. detects 6+ sessions of flat e1RM, does it say anything — or does it stay silent? Silence is a product decision with consequences.

PERSONA 7: Chloe, 25 — The Dual-Sport User

Tagline: THE PART-TIMER **Avatar:**  **Bio:** Lifts 2x/week, plays recreational soccer 2x/week, runs once a week. H.O.N. is one tool among several. Goes 3–4 days between sessions naturally — not because she’s inconsistent, but because she plays sport in between. **Traits:** multi-sport · lifts 2x/week · soccer player · runner · casual tracker · no lifting anxiety

Core tension: H.O.N. may read normal multi-sport gaps as inactivity and make her feel like a bad user when she’s actually an active one.

Test cases (at minimum 4 per phase):

Phase 1 — Home screen after 4-day gap (she played soccer twice):

- TC1.1: Opens app after 4 days. Does the home screen treat this as inactivity? What language or visual signals appear?
- TC1.2: Her e1RM trend has a 4-day gap. How does the visualization handle this — is the gap neutral or does it look alarming?
- TC1.3: She has no desire to look at charts today. She just wants to log and leave. What is the minimum viable session-start-to-log-first-set path? Count taps.
- TC1.4: She checks her last session weights to remember where she left off. Tap count and time from home screen.

Phase 2 — Monday lift after tough weekend soccer:

- TC2.1: She logs lower weights than last session (legs fatigued from soccer). Formula audit: expected e1RM for today vs. last session. App shows: _____. Does the trend line read this as regression?
- TC2.2: She selects "tired" on the feel selector. Does this affect the feedback phrase? If she's tired AND her numbers are down, does the phrase interpret the combination correctly?
- TC2.3: The feedback phrase fires on a session where numbers are down due to fatigue, not decline. Phrase audit: "you're slipping" = fail. "Off week" acknowledgment without catastrophizing = pass direction.
- TC2.4: She logs quickly and exits. Total time from app open to app close: target is under 4 minutes for a pure logging session.

Phase 3 — 10-day gap (soccer tournament):

- TC3.1: Returns after 10 days. Home screen audit — same as Wei, but she has no anxiety. Does the app treat 10 days the same as 3 weeks?
- TC3.2: Her first session back is strong — she's rested. Formula audit: e1RM may be up. Does the app correctly read rest-then-return as a positive?
- TC3.3: Rate of strength gain formula audit across the gap: does the algorithm handle a 10-day gap in a meaningful way? What happens to the weekly rate calculation?

Phase 4 — Pure utility mode (just logging):

- TC4.1: She opens the app, logs her session, and closes it in under 5 minutes with zero browsing. Is this experience smooth? Any friction points that slow her down?

- TC4.2: She never browses charts. After 3 months, what does her trend data look like — does irregular frequency affect e1RM trajectory accuracy?
- TC4.3: She has 2 sessions/week with 3–4 day gaps. Formula audit on rate of strength gain: with this cadence, does the weekly rate calculation produce meaningful results?

Design Question for Chloe: Does H.O.N. distinguish between “hasn’t trained” and “trains other things” — and if not, whose problem is that?

PERSONA 8: Kyle, 22 — The Overtrainer

Tagline: THE OBSESSIVE **Avatar:** 🔥 **Bio:** Trains 6x/week. Logs obsessively. Never takes rest days voluntarily. 6 months in. Uses H.O.N. as a scoreboard. His HRV would be tanking if he tracked it. He doesn’t know he’s overtraining. **Traits:** 6x/week · obsessive logger · no rest days · treats app as scoreboard · 6 months in · declining performance

Core tension: Does H.O.N. have a point of view on harm, or is it a neutral ledger that enables overtraining by design?

Test cases (at minimum 4 per phase):

Phase 1 — Session 6 of 6 consecutive days:

- TC1.1: He opens the app on day 6. Does anything in the app acknowledge that he’s trained 5 consecutive days?
- TC1.2: His e1RM has been declining for 2 weeks. Formula audit: verify the decline against Epley for his logged sets. Is the decline real, and does the app see it?
- TC1.3: Rate of strength gain formula audit: expected = negative value (declining e1RM). App shows: _____. Does the metric display negative rates clearly?
- TC1.4: He checks volume load. It’s climbing week over week. Formula audit: volume = sum(weight × reps). Does the chart show this clearly? Does increasing volume + declining e1RM appear anywhere together?

Phase 2 — Logging session 6:

- TC2.1: He logs his 6th session. Performance is down (lower weights or reps). Feedback phrase audit: does it call out declining trend? What exactly does it say about 2 weeks of declining e1RM?
- TC2.2: He selects “exhausted” on the feel selector for the 10th consecutive session. Does this trigger any different behavior?

- TC2.3: He tries to log his 7th day in a row. Does the app do anything different, or accept it identically to session 1?
- TC2.4: High volume + declining e1RM + exhausted feel selector — all three signals simultaneously. Does the feedback phrase integrate any of this, or address only one dimension?

Phase 3 — The design ethics test:

- TC3.1: If the app can detect: (a) 6+ consecutive days, (b) declining e1RM trend, (c) consistent exhausted feel selector — does it say anything? Should it?
- TC3.2: He reads a feedback phrase on a declining session. Does it feel neutral, concerned, or does it inadvertently praise volume?
- TC3.3: Weekly volume is 40% higher than month 1. Is there any signal that volume-without-recovery has a limit?

Phase 4 — Evening scorecard review:

- TC4.1: He browses his session history — 30 sessions in 5 weeks. Does this look healthy in the visualization or alarming?
- TC4.2: Formula audit on his overall e1RM trajectory: 6 months of data, but 2 months of decline. Does the trend line show the full arc? Does rate of strength gain reflect the turning point?

Design Question for Kyle: If H.O.N. detects 6 consecutive training days + declining e1RM + consistent exhausted feel — does it have a responsibility to say something? If not, why not?

PERSONA 9: Rachel, 32 — The Skeptic

Tagline: THE METRIC INTERROGATOR **Avatar:**  **Bio:** Data scientist. Has abandoned 6 fitness apps when she found a number that didn't add up. Trusts her spreadsheet more than any algorithm. Will use H.O.N. for exactly 2 weeks before deciding whether to commit. She will Google "Epley formula" during her first session. She is rigorous, not hostile. **Traits:** data scientist · 6 abandoned apps · spreadsheet native · formula checker · 2-week trial · rigorous

Core tension: H.O.N.'s numbers must be mathematically defensible, sourced, and consistent. One unexplained discrepancy and she leaves.

Test cases (at minimum 4 per phase):

Phase 1 — Pre-session exploration (Day 1, hasn't logged yet):

- TC1.1: She taps every "i" or info button in the app. List every explanation affordance that exists. List every metric that has no explanation.
- TC1.2: She looks for methodology documentation — how does H.O.N. calculate e1RM? Is the formula named anywhere in the app?
- TC1.3: She tries to find what "trend" means — how many sessions constitute a trend? Is this defined?
- TC1.4: She taps the feel selector before logging. What happens? Does any label explain what the emojis are for?
- TC1.5: She looks for a data export option before trusting the app with her data. Does it exist?

Phase 2 — First session formula audit (Day 1):

- TC2.1: Logs 60kg × 5. Formula audit: $Epley = 60 \times (1 + 5/30) = 70\text{kg}$. App shows: _____. Delta analysis: if 70kg, formula matches. If 70.5kg, rounding. If 71.2kg, possibly Brzycki = $60 \times (36/(37-5)) = 67.5\text{kg}$ — no, doesn't match. If 68kg, check Lander = $(100 \times 60)/(101.3 - 2.67123 \times 5) = 68.4\text{kg}$ — close match. What formula is it?
- TC2.2: Logs 60kg × 10. Formula audit: $Epley = 60 \times (1 + 10/30) = 80\text{kg}$. Brzycki = $60 \times (36/(37-10)) = 80\text{kg}$. They match at 10 reps — not a good discriminator. Now logs 60kg × 15: $Epley = 90\text{kg}$. Brzycki = $60 \times (36/(37-15)) = 98.2\text{kg}$. These diverge. App shows: _____. This is the discriminating test.
- TC2.3: She enters 60kg × 30 reps deliberately. $Epley = 60 \times (1 + 30/30) = 120\text{kg}$. This is double the weight — clearly absurd. App shows: _____. Does the app cap reps for formula purposes? Should it?
- TC2.4: She reads the feedback phrase. She tries to determine: is this phrase generated from her data or is it a template? She logs an identical session the next day with the same exact weights and reps. Do the phrases differ? Should they?
- TC2.5: She verifies that rate of strength gain after session 1 = 0 or undefined (only 1 data point). App shows: _____. If it shows any non-zero value, that's a bug.

Phase 3 — Week 1 formula verification:

- TC3.1: After 5 sessions, she plots her e1RM. She manually calculates all 5 values using Epley. Formula audit: list all 5 expected vs. observed values. Any discrepancy triggers investigation.
- TC3.2: Rate of strength gain formula audit: she calculates manually = $(\text{session 5 e1RM} - \text{session 1 e1RM}) \div (\text{days elapsed} \div 7)$. App shows: _____. Does it match?

- TC3.3: The app shows an "upward trend." She counts the data points supporting this claim. Is it 2 sessions? 3? She considers: is 2 sessions enough to call something a trend? Does the app define this threshold?
- TC3.4: She deliberately logs a session with identical data to session 1. Does e1RM match? Does rate of strength gain update correctly?

Phase 4 — Week 2 trust decision:

- TC4.1: She found a discrepancy — her manual Epley and the app's e1RM differ by 2.5kg. She has no way to explain it. Is there any in-app path to understanding this?
- TC4.2: She tries again to export her data. Still no export? This is her last week.
- TC4.3: She tries to reverse-engineer the formula from the data she has. Given her 10 logged sessions, can she determine which formula the app uses from the outputs alone?

Design Question for Rachel: If a data scientist cannot verify H.O.N.'s formula from within the app, will she trust it with 6 months of her training data?

PERSONA 10: Tom, 28 — The Breaker

Tagline: THE STRESS TESTER **Avatar:**  **Bio:** QA engineer by profession. Downloaded H.O.N. to find what breaks when you use it wrong. Has broken 3 apps this month just by using them normally but incorrectly. Methodical. He will find your edge cases. Not malicious — just thorough. **Traits:** QA engineer · edge case hunter · methodical · non-lifter · adversarial by nature · thorough

Core tension: Every untested input path is a bug waiting to happen. Tom walks all of them.

Test cases (at minimum 4 per phase):

Phase 1 — Input boundary testing:

- TC1.1: Enters 0kg weight. Does the app accept it? Log it? Calculate e1RM? Formula audit: Epley with 0kg = 0 regardless of reps. Does app show 0?
- TC1.2: Enters 999kg weight, 1 rep. Formula audit: Epley = $999 \times (1 + 1/30) = 1032.3\text{kg}$. App shows: _____. Does this appear in his trend? Does it corrupt the y-axis scale?
- TC1.3: Enters 60kg, 100 reps. Formula audit: Epley = $60 \times (1 + 100/30) = 260\text{kg}$. This is absurd. App shows: _____. Does the app accept 100 reps? Is there a rep cap? Should there be?
- TC1.4: Enters 60kg, 0 reps. What happens? Is 0 reps a valid set?

- TC1.5: Enters negative weight: -10kg. What happens?
- TC1.6: Enters decimal weight: 82.5kg. Does the input support this? What about 77.5kg and 92.5kg?
- TC1.7: Enters a weight with a comma instead of decimal: "82,5". What happens?

Phase 2 — Navigation abuse:

- TC2.1: Starts a session. Immediately taps back. Is there a "discard session?" confirmation? Or silent discard?
- TC2.2: Starts a session, logs 5 sets, taps back. Same question — what happens to the data?
- TC2.3: Taps "log set" 15 times in 3 seconds. How many sets are logged? Is there rate limiting?
- TC2.4: Opens app and immediately force-quits during the splash screen animation. Reopens. Is anything broken?
- TC2.5: Starts a session. Backgrounded for 30 minutes. Comes back. Session still active?
- TC2.6: Logs a set, then immediately undoes it (if undo exists). Then logs it again. Does the history show 1 or 2 sets?

Phase 3 — Data state edge cases:

- TC3.1: Logs a session with 1 set of 1 exercise. Valid session? Appears in history? e1RM calculated from single point? Formula audit: single set e1RM vs. displayed.
- TC3.2: Logs 15 different exercises, 10 sets each (150 total sets). Does the UI degrade? Does the feedback phrase handle this? What does the session summary look like?
- TC3.3: Logs a session, ends it, immediately reopens the completed session (if editable), adds 5 more sets, ends again. How does this affect e1RM and trend calculations? Formula audit on the final state.
- TC3.4: Uses airplane mode throughout an entire session. Logs 10 sets. Ends session. Reconnects. Is session in history? Formula audit — does the offline session produce the same e1RM as if logged online?
- TC3.5: Logs two sessions on the same day. Is this valid? Do both appear in history? Does the trend line handle same-day double sessions?

Phase 4 — Feel selector abuse:

- TC4.1: Skips the feel selector for 20 consecutive sessions. Does anything in the app

respond to having no feel data?

- TC4.2: Selects every emoji option in sequence (if multi-select is possible) on the same session. What's the final recorded state?
- TC4.3: Force-quits immediately after the feel selector appears but before selecting. What is the saved feel state?

Design Question for Tom: How many of these inputs were tested during development — and which ones produce unexpected behavior that would affect real users who never intended to break anything?

PERSONA 11: Alex, 26 — The Vanity Logger

Tagline: THE CHERRY PICKER **Avatar:** ✨ **Bio:** Logs every session where they hit a PR or feel strong. Skips sessions where they underperform. Has done this for 3 months. Their H.O.N. history is systematically optimistic. They don't realize this is a problem. **Traits:** selective logger · PR-only habit · 3 months in · unconsciously biased · progress-motivated · doesn't see the gap

Core tension: H.O.N. is building a trend from cherry-picked data and doesn't know it. The trend line is accurate to logged sessions — which are not representative of actual training.

Test cases (at minimum 4 per phase):

Phase 1 — The good week:

- TC1.1: Logs a strong bench session (80kg × 6). Formula audit: $\text{expected e1RM} = 80 \times (1 + 6/30) = 96\text{kg}$. App shows: _____. Trend is rising.
- TC1.2: Two days later, logs a great squat session. Checks trend. Everything looks like progress.
- TC1.3: Feedback phrase fires after a good session. Phrase audit: does it praise "consistent progress" when the consistency is fabricated by selective logging?
- TC1.4: He checks his session frequency. 3 sessions logged in 2 weeks. The app may read this as normal frequency. What is the actual training frequency he's not logging?

Phase 2 — The bad week (selective non-logging):

- TC2.1: Monday: bench is heavy, only hits 3 reps instead of 5. He doesn't log. Gap appears in history. Does the app notice or react to the Monday gap?
- TC2.2: Wednesday: squats feel off. He logs it under "back squat" instead of "squat" to

prevent the bad session from affecting his “squat” trend. Does H.O.N. allow this? Are “squat” and “back squat” treated as the same exercise or different?

- TC2.3: Formula audit on the Wednesday session logged under the different name: expected $e1RM$ = same formula, different history chain. How does this appear in the trend?
- TC2.4: Friday: great deadlift. He logs it. His history now shows: nothing Mon, nothing Wed, good Fri. Does the 2-day gap register as anything in the visualization?

Phase 3 — The survivorship bias problem:

- TC3.1: After 3 months of selective logging, Alex’s trend line looks like perfect linear progress. Formula audit: pick 5 plotted $e1RM$ values. Verify against Epley for the underlying sets. Are they accurate? They are — but they represent only his best sessions.
- TC3.2: The feedback phrase calls his progress “consistent upward trend.” This is true for the data logged. Is it possible to game the phrase bank by only logging good days?
- TC3.3: He has never logged a session where he got fewer reps than last time. Does the phrase bank’s “reps outcome” tier ever fire its lowest tier for him? Should it?
- TC3.4: Rate of strength gain formula audit: his rate looks impressive. But the denominator (weeks elapsed) includes weeks where bad sessions went unlogged. Does the formula account for data gaps, or assume logged sessions represent complete training history?

Phase 4 — The honest log:

- TC4.1: He has an exceptionally bad session and decides to log it. He logs bench $72\text{kg} \times 3$ (vs. usual $80\text{kg} \times 6$). Formula audit: expected $e1RM = 72 \times (1 + 3/30) = 79.2\text{kg}$ vs. previous 96kg . A major drop. Feedback phrase audit: does it feel fair or punishing?
- TC4.2: The trend line now has a visible cliff. Does the visualization make this feel like information or shame?
- TC4.3: Rate of strength gain formula audit: the drop from 96kg to 79.2kg changes his rate significantly. Does the weekly rate now show negative? Does the app communicate this clearly?
- TC4.4: He reads the phrase after the honest bad session. If it’s too harsh, he won’t log bad days again. If it’s too soft, he gets no signal. What’s the right calibration?

Design Question for Alex: Can H.O.N.’s trend algorithm detect that it’s only seeing a user’s best sessions — and if it can’t, is it misleading them about their own progress?

PERSONA 12: Sam, 24 — The First Logger

Tagline: THE COLD START **Avatar:** 🧑 **Bio:** Just downloaded H.O.N. 4 minutes ago. Has never used a workout tracking app. No prior data. No history. No trends. Every screen is a first impression. Every label is either self-evident or confusing. Every empty state is either welcoming or alienating. **Traits:** brand new user · zero history · no app experience · first session ever · every assumption untested · high churn risk

Core tension: The app was designed by someone who knows what it does. Sam has no idea. Every assumption baked into the UI is being tested right now.

Test cases — run chronologically, minute by minute:

Minutes 0–2: Splash and first screen:

- TC1.1: Splash animation plays (double amber rules → HON → subtitle → dots → tagline “Show up. That’s enough.”). What does Sam understand about the app’s purpose from this animation alone?
- TC1.2: First screen after splash: what is on it? Does Sam know what to do? Time how long before they take any action.
- TC1.3: Every piece of text on the first screen — is it self-evident without context? List every term that assumes prior knowledge.

Minutes 2–5: Exploration:

- TC2.1: Sam taps around without a goal. What is the first thing that confuses them? What do they tap that doesn’t do what they expected?
- TC2.2: They see “e1RM” somewhere. They don’t know what it means. What happens if they tap it? Is there any explanation?
- TC2.3: They find the feel selector before logging anything. They tap it. What happens? Do the emojis have labels?
- TC2.4: They look for a tutorial or “how to start.” Does one exist? If not, what do they do?

Minutes 5–10: First session attempt:

- TC3.1: They tap whatever starts a session. How many screens before they can log their first set? Count every screen.
- TC3.2: They type “bench” in exercise search. They pick “Barbell Bench Press.” Is this the right choice? Do they know? Is there anything to help them decide?

- TC3.3: They see weight and rep inputs. They haven't thought about kg vs. lbs. What unit is the app in? Is this visible before they enter data? Can they change it?
- TC3.4: They enter 40kg × 8 and log the set. What happens immediately after? Is there any confirmation? Does the e1RM appear? Formula audit: $\text{expected} = 40 \times (1 + 8/30) = 50.7\text{kg}$. App shows: _____. Does Sam understand this number?

Minutes 10–20: Navigating mid-session:

- TC4.1: The rest timer appears. Sam doesn't know what to do with it. Is it clear what they're waiting for? Can they skip it?
- TC4.2: They try to add a second exercise. Can they find how to do this without help?
- TC4.3: They accidentally log the same exercise twice. Can they delete or merge the duplicate?
- TC4.4: They want to see what they've logged so far without ending the session. Is there a summary view mid-session?

Minutes 20–25: Ending the session:

- TC5.1: They find "end session." How many taps from anywhere in the session?
- TC5.2: Feel selector appears. They've never seen this before. Do they know what to tap? Do the emojis have labels or tooltips?
- TC5.3: Feedback phrase appears. This is the first thing H.O.N. has ever said to them. Full phrase audit: does it reference their data (40kg, 8 reps)? Does it use the word "e1RM" without explanation? Does it make them feel like the session counted?
- TC5.4: Session summary appears. List every element they see. Mark each one: understood immediately / confused / ignored.

Minutes 25–60: After the session:

- TC6.1: They close the app. 30 minutes later they reopen it. Does the home screen feel different now that one session exists? Does it acknowledge them?
- TC6.2: They try to find their session. Tap count from home screen to the session detail.
- TC6.3: They see e1RM in their session summary. Formula audit: does the value match Epley? Do they understand what the number means without explanation?
- TC6.4: They have one data point. No trends possible. What does the app show them where a trend would eventually appear? Is the empty state honest and inviting?

The return question — display this prominently at the end of Sam's section, large

typography, no scorecard:

“Would Sam open H.O.N. again tomorrow?” “And if so — what would make them?”

Design Question for Sam: What is the single most important thing the first screen must communicate — and does it currently communicate it?

PART 4: NAVIGATION EFFICIENCY AUDIT

This is a separate tab, not a persona. Design it as an interactive audit tool — the developer sits with their phone and works through it physically.

A. The 8 Core Journeys

For each journey, display:

- Goal statement
- Expected tap sequence (numbered steps)
- Tap count badge
- Severity:  Efficient (1–3 taps) ·  Acceptable (4–5 taps) ·  Needs Work (6+ taps)
- “Fastest possible path” — what would it take to reduce this to 1–2 taps?
- A blank “Actual tap count on my device:” field for the developer to fill in

Journey 1: From home screen, find the weight used for bench press in the last session.

Journey 2: From home screen, start a new session. **Journey 3:** From active session, log a

new set for an exercise already in progress. **Journey 4:** From active session, switch to a

new exercise. **Journey 5:** Fix a weight entered wrong — immediately after logging. **Journey**

6: End session → feel selector → read feedback phrase. **Journey 7:** From anywhere, view

e1RM trend for bench press over 6 weeks. **Journey 8:** Navigate to a session from 3 weeks

ago.

B. Thumb Zone Diagram

Draw an iPhone (portrait, rounded corners) in pure HTML/CSS — no images. Divide it into three zones:

-  Natural reach — bottom 40%, center-weighted (comfortable thumb zone)
-  Stretch zone — middle 40% (requires extension)

-  Dead zone — top 20% (requires grip shift or second hand)

Below the diagram, audit these specific H.O.N. elements by zone:

- Log set button
- Weight input field
- Rep input field
- Rest timer display
- End session button
- Exercise search
- Back navigation
- Feel selector

For each: note which zone it likely falls in, and whether that's appropriate for how often it's used.

C. One-Handed Use Checklist

10 items. Developer performs each with right thumb only, no repositioning. Each item: 

Pass /  Fail /  Size-dependent.

1. Tap "log set" from standard logging position
2. Enter a 3-digit weight (e.g. 100) on the numpad
3. Enter a 2-digit rep count (e.g. 12)
4. Reach and tap the back/cancel button
5. Start a new exercise mid-session
6. Scroll through exercise search results
7. Tap to end the session
8. Select a feel emoji
9. Dismiss the rest timer
10. Read the feedback phrase without repositioning

Note: iPhone 15 Pro Max users have 8–10% more screen real estate in the dead zone vs. iPhone SE users. Test on the smallest device you support.

D. Speed Benchmarks

Interaction	Target time	Failure threshold	My actual time
Log a set (weight pre-filled from last set)	<5 sec	>10 sec	—
Start a new session from home	<4 sec	>8 sec	—
Find last session's bench weight	<8 sec	>15 sec	—
End session + through feel selector	<10 sec	>20 sec	—
Navigate to a session from 3 weeks ago	<12 sec	>25 sec	—
View e1RM trend for one exercise	<10 sec	>20 sec	—

Test all of these immediately after a heavy set, with your non-dominant hand. That is the real benchmark.

PART 5: PRE-LAUNCH AGGREGATION TAB

Title: 🚫 **Pre-Launch Master View**

Aggregate all issues from all 12 personas plus the navigation audit into three tiers. For each issue:

- Severity badge
- Persona tag(s)
- Issue title
- Why it matters (2–3 sentences specific to H.O.N.)
- How to test it (specific physical steps on device)

Ship-Blockers (must resolve before any external user):

At minimum include:

- Input validation failures that corrupt trend data (Tom: 100 reps produces 260kg e1RM via Epley — appears in trend, corrupts y-axis scale)
- Formula discrepancy with no in-app explanation (Rachel: manual Epley $\neq$ app display, no methodology documented)
- Offline session data loss (Tom: airplane mode session — must save locally and sync)
- Unit setting (kg vs. lbs) with no default confirmation (Sam: first session logged in wrong units, data permanently wrong)
- Session state after force-quit (Marcus, Tom: orphaned session with no recovery path)
- e1RM calculation with 1 rep (Serena, Tom: Epley with 1 rep produces $\text{weight} \times 1.033$ — is this capped or validated?)

Fix Before Launch (will drive week-one churn):

At minimum include:

- Decimal weight input (82.5kg, 92.5kg) — affects all serious lifters
- Navigation depth >3 taps to last-session data (structural, appears across 7 personas)
- Rapid-tap protection on “log set” (Tom: 15 identical sets in 3 seconds)
- Zero explanation affordance for e1RM, rate of strength gain, trend direction, INOL
- Feel selector — is it functional or decorative? Must be one or the other, clearly
- Bad-day logging phrase quality — neutral tone is non-negotiable
- Exercise disambiguation for beginners (Danny, Sam, Alex: wrong exercise selected, history fragmented)
- Feedback phrase with 0–2 sessions (Sam, Danny: cold-start phrase floor must be specific, not generic)
- Retroactive session timestamp (Serena: session logged 45 min after it occurred shows wrong time)
- Rep cap for formula purposes (Tom: 100 reps should trigger validation, not 260kg e1RM)

Post-Launch (survivable for v1):

- Data export / CSV
- Methodology documentation in-app (“H.O.N. uses the Epley formula: $\text{weight} \times (1 +$

reps/30)")

- Gap annotation (Wei, Alex: ability to mark a gap period with context)
 - Trend minimum data threshold label ("trend shown after 3 sessions")
 - Rest timer dismissal/hide for retroactive loggers (Serena)
 - Stale trend warning for Rachel's 2-data-point "upward trajectory"
 - Multi-exercise comparison view (James: bench vs. squat side by side)
 - Session frequency vs. inactivity distinction (Chloe: 4-day gap = normal for her)
-

PART 6: CROSS-PERSONA PATTERNS

Eight structural patterns that span multiple personas. These are product decisions, not isolated bugs.

Pattern A — The feedback phrase is load-bearing for every persona, for different reasons Marcus needs retention. Priya needs credibility. Danny needs self-worth. Wei needs compassion. James needs honesty about plateaus. Kyle needs a signal he's overreaching. Alex needs accuracy about cherry-picked data. Sam needs a first impression that makes him come back. The phrase must do all of this from the same phrase bank architecture.

Pattern B — The app is always used in physical conditions Sweaty hands (Marcus, Priya, Danny). One-handed (Marcus, Priya, Chloe). Standing and moving (all). Chalky or gloved fingers (Serena). The design review happened at a desk. The real test happens at the gym.

Pattern C — History access is the second-most-used feature Every persona checks prior session data before or during a session. If this takes more than 3 taps, users remember from memory instead — and memory is inaccurate. This degrades data quality silently and compounds over months.

Pattern D — The formula is the foundation of everything e1RM drives trend. Trend drives the feedback phrase. The feedback phrase drives retention. If the formula is wrong, inconsistent, or undocumented, every downstream output is compromised. Rachel and Tom surface this from opposite directions — she tests precision, he tests boundaries. Both expose the same gap.

Pattern E — The cold-start experience affects more users than any other Sam represents every new user. Every other persona has prior context. Sam has none. And every user was Sam once. The first 5 minutes of H.O.N. determine whether the rest of this

simulation is relevant for a given person — because if they leave in minute 3, none of the other features matter.

Pattern F — Input validation is invisible until it isn't Tom deliberately enters absurd values. But Alex accidentally fragments history by using a slightly different exercise name. Serena needs decimal weights. Danny doesn't know what to search for. All of these are input problems — just wearing different clothes. Validation is not just about rejecting bad data; it's about guiding users toward good data.

Pattern G — The app has an implicit behavioural opinion whether it intends to or not Kyle is rewarded for 6x/week with the same logging experience as 3x/week. Alex is rewarded for selective logging with a "consistent progress" phrase. Chloe is potentially penalized for multi-sport gaps. H.O.N.'s silence on behaviour IS a behaviour design. Every neutral choice is still a choice.

Pattern H — First 3 sessions are the highest-churn window for every beginner-adjacent persona Sam (session 1), Danny (sessions 1–3), Wei (return sessions 1–2), Chloe (irregular frequency early on). The phrase bank's floor experience — zero trend data, minimal history — must be designed explicitly. It is not an edge case. It is the default state for weeks.

PART 7: PHYSICAL TEST PROTOCOL

Title: **Do These Before Shipping. With Your Body. On Your Device.**

15 steps. Each one is something the developer does physically:

1. **The Epley Audit.** Log $60\text{kg} \times 5$. Write down what the app shows. Calculate manually: $60 \times (1 + 5/30) = 70\text{kg}$. Do they match? Then log $60\text{kg} \times 15$: Epley = 90kg , Brzycki = 98.2kg — these diverge. Which does the app use?
2. **The Boundary Test.** Enter 0 reps. Enter 999kg. Enter 100 reps. Enter 82.5kg. Enter -10kg . For each: accept, reject cleanly, or break?
3. **The Rapid Tap Test.** Find "log set." Tap it 10 times in 3 seconds without changing values. Count the logged sets.
4. **The Airplane Mode Test.** Airplane mode on. Start a session. Log 4 sets. End session. Reconnect. Find the session in history.
5. **The Interruption Test.** Start a session. Log 2 sets. Background the app. Wait 7 minutes. Let the screen lock. Return. Is everything exactly where you left it?

6. **The Force-Quit Test.** Mid-session, force-quit from app switcher. Reopen. What do you see?
 7. **The Sweaty Hands Test.** Do a set of something. Don't wipe your hands. Immediately attempt to log the set. Note every friction point.
 8. **The Stranger Test.** Hand your phone to someone who has never seen H.O.N. Ask them to start a session and log one set. No instructions. Watch silently. Note every hesitation.
 9. **The Bad Day Test.** Deliberately enter weights lower than your last session. Read the feedback phrase out loud to someone. Does it feel fair?
 10. **The 4-Day Gap Test.** Don't open H.O.N. for 4 days. Open it as if you just woke up. Read the home screen as a stranger. What does it tell you?
 11. **The Navigation Timer Test.** From home screen, time: (a) find last bench weight, (b) start new session, (c) find a session from 3 weeks ago. Write the times down.
 12. **The Cherry Pick Test.** Log only strong sessions for 2 weeks. Read your trend. Does H.O.N. know it's only seeing your good days?
 13. **The One-Handed Test.** Do the entire logging flow — start session, pick exercise, enter weight and reps, log set — with your non-dominant hand only. Note every moment you need your other hand.
 14. **The Formula Discrimination Test.** Log 60kg × 30 reps. What does the app show? (Epley = 120kg, double the weight. This should never appear in a meaningful trend.) Now log 60kg × 1. What does it show? (Epley = 62kg — is a 1-rep set treated differently?)
 15. **The Return Question Test.** Delete the app. Reinstall. Log 2 sessions as a beginner. Read every piece of text out loud. Would you come back on day 3?
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PART 8: OUTPUT REQUIREMENTS

File

Single self-contained HTML file. File name: `hon-ux-simulation-complete.html` . No external dependencies except Google Fonts. No frameworks. Vanilla HTML, CSS, JavaScript only.

Navigation tabs (top, sticky)

- 01 Marcus · 02 Priya · 03 Danny · 04 Wei · 05 Serena · 06 James · 07 Chloe · 08 Kyle · 09

Per-persona structure

1. Persona card (emoji, name, age, tagline, bio, trait pills)
2. 5 phases, each with labeled test cases (TC X.X format)
3. Formula audit blocks wherever calculations appear
4. Annotations (Friction / Feeling / Interruption / Flow Check / Formula Audit)
5. Scorecard
6. Single "Design Question"

Formula Audit Block style

Distinct card within annotations. Lavender color scheme. Shows:

- Inputs used
- Formula written out with actual numbers
- Expected output
- "App shows: ____" (blank, developer fills in)
- Delta analysis (pre-written interpretations of possible discrepancies, including which alternative formula would produce each result)

Sam's section

Ends with the return question in large display typography. No scorecard.

Design system — enforce throughout

Exact colors as specified. DM Serif Display for headings and persona names. DM Mono for all labels, badges, timestamps, test case numbers, formula text. DM Sans for narrative body text and annotations. Dark backgrounds throughout. Amber reserved strictly for hero metric.

Quality bar

Every test case must make the developer think: "I haven't done this yet." Every formula audit

block must give the developer enough information to investigate a discrepancy without leaving the document. Every narrative moment must describe a real human in a real physical situation — not an abstract scenario. The document must be usable as a physical QA checklist the developer works through with their phone in hand.